

Cyclic TSA-mIHC/IF Staining Protocol

for detecting COX-2, PTGER4 and PIK3CA in human adenoma tissues

1. Deparaffinization
 - 1.1. Immerse slides in xylene three times for 15 minutes each.
 - 1.2. Immerse slides in 100% alcohol three times for 5 minutes each.
 - 1.3. Immerse slides in 95% alcohol once for 5 minutes.
 - 1.4. Rinse in deionized water, then transfer to fresh phosphate-buffered saline (PBS) buffer (pH7.2) for 5 minutes.
2. Antigen retrieval

Perform antigen retrieval using EDTA buffer (pH8.0) in a pressure cooker. Rinse in PBS for 5 minutes, drain the slides, and wipe off excess buffer.
3. Blockings
 - 3.1. Incubate with 3% Hydrogen Peroxide Solution (H_2O_2) for 5 minutes, then rinse in PBS for 5 minutes.
 - 3.2. Incubate with 10% normal goat serum for 15 minutes at room temperature (RT); do not rinse.
4. First Biomarker labeling: PTGER4 (CF594)
 - 4.1. Mouse anti-PTGER4 (clone 4A2A12, Proteintech, Cat# 66921-1-Ig, 1.2 mg/mL), dilution 1:1200 (1:200 for standard IHC), incubate for 1 hr at RT.
 - 4.2. Goat anti-mouse IgG HRP polymer (abcam, Cat# ab214879), 30 minutes at RT.
 - 4.3. CF594-TSA (Biotium, Cat#92174), 1:400 dilution in Tyramide Amplification Buffer Plus (TAB Plus, Biotium, Cat# 22027), 10 minutes at RT.

Note 1: For CF dyes (0.5mg), dissolve in 1mL DMSO to make stock solution (stored at -20°C for up to 1 year). Working solution: 1:100-250 (2-5ug/mL) using TAB Plus.

Note 2: To prepare for the working amplification buffer, add 1 uL of 30% H_2O_2 to 200 uL of 1X Tyramide Plus Buffer and mix well to make a 0.15% H_2O_2 solution (stored at 4°C). Add 1 uL of the 0.15% H_2O_2 solution to 100 uL TAB Plus solution, for a final concentration of 0.0015% H_2O_2 in working TAB Plus.

Note 3: After each step of antibody labeling, gently rinse slides with PBS using a wash bottle and place slides in PBS buffer bath three times for 5 minutes each.
5. Second biomarker labeling: PIK3CA (CF680)
 - 5.1. Repeat blocking steps (3a and 3b).
 - 5.2. Rabbit anti-PIK3CA (Clone SP139), dilution 1:1000 (1:50 for standard IHC), incubate overnight at 4°C.

- 5.3. Goat anti-rabbit IgG HRP polymer (abcam, Cat# ab214880), 30 minutes at RT.
- 5.4. CF680R-TSA (Biotium, Cat#92196), 1:200 dilution in TAB Plus, 10 minutes at RT.
6. Microwave treatment
Place slides in citrate buffer (pH 6.0), set high power for 135s, then low power for 1500s.
Microwave information: Model BP-090, Research & Applications, Inc.,
<https://microvaveresearch.com/BP-090.php>
7. Third biomarker labeling: COX2 (CF488A)
 - 7.1. Repeat blocking steps (3a and 3b).
 - 7.2. Rabbit anti-COX-2 (Clone D5H5, Cell Signaling, Cat# 12282), dilution 1: 300 (1:50 for standard IHC), incubate 1 hr at RT.
 - 7.3. Goat anti-Rabbit IgG HRP polymer, 30 minutes at RT.
 - 7.4. CF488A-TSA (Biotium, Cat#92172), 1:200 dilution in TAB Plus, 10 minutes at RT.
8. Fourth biomarker labeling: Pan CK (Cy7)
 - 8.1. Repeat blocking steps (3a and 3b).
 - 8.2. Mouse anti-Pan CK (AE1/AE3, Novusbio, Cat# NBP2-29429), dilution 1:300, incubate overnight at 4°C.
 - 8.3. Goat anti-mouse IgG HRP polymer (abcam, Cat# ab214879), 30 minutes at RT.
 - 8.4. Cy7-TSA (AAT Bioquest, Cat# 11064), 1:200 dilution in TAB Plus, 10 minutes at RT.
9. Nuclear counterstain
DAPI (1mg/ml stock solution), 1:1000 dilution, incubate for 10 minutes at RT.
Rinse in PBS for 5 minutes.
10. Mounting and imaging
Coverslip slides using either 50% glycerol for temporary mounting followed by cyclic mIHC/IF staining or Fluoromount-G™ Mounting Medium without DAPI (Invitrogen, Cat# 00-4958-02) for long-term storage. Store slides at 4°C in a covered slide box. Acquire fluorescence images within one week using the Olympus BX61 Trinocular Fluorescence Motorized Upright Microscope equipped with six fluorescence filter sets.
- 11. If multiple round staining is needed for detecting additional biomarkers**
 - 11.1. 4% Paraformaldehyde (PFA) incubates for 30 minutes in the humidity chamber at room temperature. Wash in ddH₂O and rinse in PBS for 5 minutes.
 - 11.2. Microwave treatment (as Step6) for stripping antibodies.
 - 11.3. Fluorescence quenching

- 11.3.1. Stock solutions were 1M sodium bicarbonate (NaHCO_3 , FW 84.01) and 30% hydrogen peroxide. The NaHCO_3 solution was adjusted to pH 10 using sodium hydroxide.
- 11.3.2. The fresh working solution was made by mixing 1 volume of 1 M NaHCO_3 , 3 volumes of distilled water, and 1 volume of 30% hydrogen peroxide.
- 11.3.3. Slides were placed in the bleaching solution for 3 hours at RT under a table lamp, followed by a thorough washing with PBS.
- 11.4. After staining and image acquisition of the second 5-plex mIHC/IF staining cycle, repeat steps 11.1 – 11.3 for the third staining cycle.